

Name _____ Date _____ Period _____

Directions. Follow the six steps on your guide for every problem. Draw and label a diagram first, write your givens as derivatives, and **differentiate before you plug in**. Give each answer with units, and say whether the quantity is increasing or decreasing.

PART I · WARM-UP One equation, straight from the formula bank

1 A stone dropped into a still pond sends out a circular ripple. The radius of the ripple grows at 2 ft/s. How fast is the area inside the ripple increasing when the radius is 10 ft?

GIVEN	FIND	EQUATION
_____	_____	_____

Answer

2 The edges of a cube are expanding at 0.5 cm/s. How fast is the volume of the cube increasing when each edge is 4 cm long?

GIVEN	FIND	EQUATION
_____	_____	_____

Answer

3 The length of a rectangle is increasing at 3 cm/s while its width is decreasing at 2 cm/s. At the instant the length is 12 cm and the width is 5 cm, how fast is the area changing? Is the area increasing or decreasing?

GIVEN	FIND	EQUATION

Answer

4 Water is poured into a cylindrical tank with a radius of 3 m at a rate of 36 m³/min. How fast is the water level rising when the water is 2 m deep?

GIVEN	FIND	EQUATION

Answer

PART II • CHALLENGE Draw the diagram first. Some need similar triangles or trig.

5 A 5-ft-tall woman walks *toward* a 20-ft streetlight at 4 ft/s. When she is 10 ft from the pole:

- (a) How fast is the length of her shadow changing? Is it getting longer or shorter?
(b) How fast is the tip of her shadow moving?

YOUR DIAGRAM

(a)

(b)

6 A 13-ft ladder leans against a wall. The *top* of the ladder slides down the wall at 2 ft/s. When the top is 5 ft above the ground:

- (a) How fast is the bottom of the ladder sliding away from the wall?
(b) How fast is the angle between the ladder and the ground changing?

YOUR DIAGRAM

(a)

(b)

- 7** A boat is pulled toward a dock by a rope tied to its bow. The rope runs through a pulley on the dock that sits 6 ft higher than the bow, and the rope is pulled in at 2 ft/s.
- (a) How fast is the boat approaching the dock when it is 8 ft from the dock?
- (b) Answer the same question when the boat is 4.5 ft from the dock. What do you notice?

YOUR DIAGRAM

(a)

(b)

- 8** A baseball diamond is a square with 90 ft between bases. A batter runs from home plate toward first base at 24 ft/s. When he is halfway to first base:
- (a) How fast is his distance to second base changing?
- (b) How fast is his distance to third base changing?

YOUR DIAGRAM

(a)

(b)